

Setting Up LEMP on Ubuntu with Nginx

Introduction

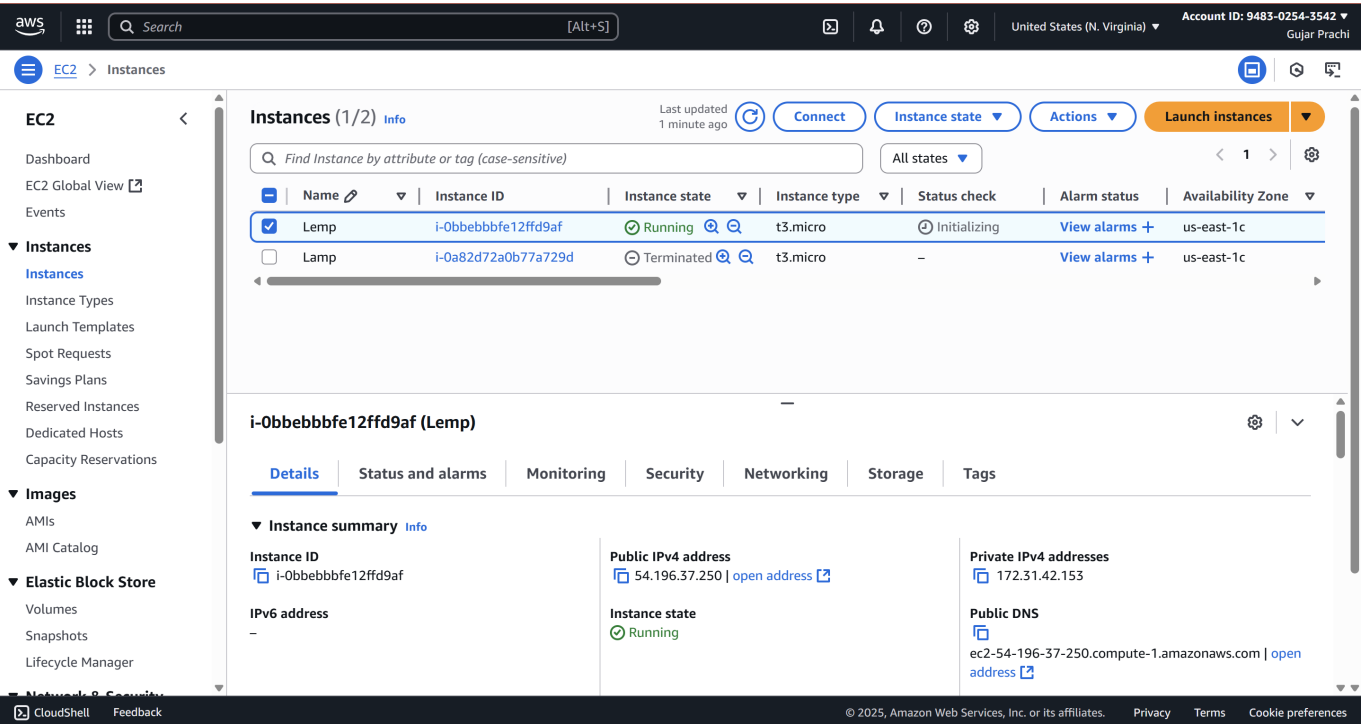
This project explains how to set up a LEMP stack (Linux, Nginx, MySQL/MariaDB, and PHP) on Ubuntu. It covers installing and configuring each component to serve dynamic websites efficiently using Nginx as the web server. By following these steps, you'll have a fully functional environment for hosting PHP-based applications.

Prerequisites

- AWS account (EC2 access)
- Basic Linux knowledge
- SSH key pair
- Security group: ports 80 (HTTP), 22 (SSH)
- Ubuntu EC2 instance
- Basic knowledge of Nginx, MySQL, PHP

Steps to Deploy LEMP Stack

Step 1: Login into AWS Account and Launch Ec2 Instance



Step 2: Establish a secure SSH connection to your EC2 instance and update the system packages.

- Copy the SSH connection command from the SSH client and paste it into Git Bash to connect to the instance.

```
# to update
sudo apt update
```

```
ubuntu@ip-172-31-42-153: ~
prachi@Admin MINGW64 /c/Prachi_workspace/ssh-key
$ ssh -i "keypair.pem" ubuntu@ec2-54-196-37-250.compute-1.amazonaws.com
The authenticity of host 'ec2-54-196-37-250.compute-1.amazonaws.com (54.196.37.250)' can't be established.
ED25519 key fingerprint is SHA256:SRZbofGh+qQBokSPFIUmZr1h8g01sfZlFvYq03Y0ay.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-54-196-37-250.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1011-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Aug 23 19:31:02 UTC 2025

System load:  0.02           Temperature:   -273.1 C
Usage of /:   25.6% of 6.71GB Processes:      115
Memory usage: 24%           Users logged in: 0
Swap usage:   0%            IPv4 address for ens5: 172.31.42.153

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-42-153:~$ sudo apt update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1083 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.6 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [881 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [195 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [52.3 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [17.0 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:15 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [18.5 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [4288 B]
Get:17 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:18 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [380 B]
Get:19 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:20 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:21 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:22 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 components [35.0 kB]
```

Step 3: Install the Nginx webserver.

```
# to install nginx
sudo apt install nginx -y
```

```
ubuntu@ip-172-31-42-153: ~
ubuntu@ip-172-31-42-153:~$ sudo apt install nginx -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  nginx-common
Suggested packages:
  fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  nginx nginx-common
0 upgraded, 2 newly installed, 0 to remove and 5 not upgraded.
Need to get 564 kB of archives.
After this operation, 1596 kB of additional disk space will be used.
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx-common all 1.24.0-2ubuntu7.4 [43.4 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx amd64 1.24.0-2ubuntu7.4 [521 kB]
Fetched 564 kB in 0s (18.5 MB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database ... 71718 files and directories currently installed.)
Preparing to unpack .../nginx-common_1.24.0-2ubuntu7.4_all.deb ...
Unpacking nginx-common (1.24.0-2ubuntu7.4) ...
Selecting previously unselected package nginx.
Preparing to unpack .../nginx_1.24.0-2ubuntu7.4_amd64.deb ...
Unpacking nginx (1.24.0-2ubuntu7.4) ...
Setting up nginx-common (1.24.0-2ubuntu7.4) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
Setting up nginx (1.24.0-2ubuntu7.4) ...
* Upgrading binary nginx
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for ufw (0.36.2-6) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-42-153:~$ ]
```

Step 4: Install MySQL.

```
# to install mysql
sudo apt install mysql-server
```

```
ubuntu@ip-172-31-42-153: ~
ubuntu@ip-172-31-42-153:~$ sudo apt install mysql-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libbcbi-fast-perl libbcbi-pm-perl libclone-perl libcode-locale-perl libevent-pthreads-2.1-7t64 libfcgi-bin libfcgi-perl libfcgi0t64 libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl
  libhttp-date-perl libhttp-message-perl libio-html-perl liblwp-mediatypes-perl libmecab2 libprotobuf-lite32t64 libtimedate-perl liburi-perl mecab-ipadic mecab-ipadic-utf8 mecab-utils mysql-client-8.0
  mysql-client-core-8.0 mysql-common mysql-server mysql-server-8.0 mysql-server-core-8.0
Suggested packages:
  libdata-dump-perl libipc-sharedcache-perl libio-compress-brotli-perl libbusiness-isbn-perl libregexp-ipv6-perl libwww-perl mailx tinyca
The following NEW packages will be installed:
  libbcbi-fast-perl libbcbi-pm-perl libclone-perl libcode-locale-perl libevent-pthreads-2.1-7t64 libfcgi-bin libfcgi-perl libfcgi0t64 libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl
  libhttp-date-perl libhttp-message-perl libio-html-perl liblwp-mediatypes-perl libmecab2 libprotobuf-lite32t64 libtimedate-perl liburi-perl mecab-ipadic mecab-ipadic-utf8 mecab-utils mysql-client-8.0
  mysql-client-core-8.0 mysql-common mysql-server mysql-server-8.0 mysql-server-core-8.0
0 upgraded, 28 newly installed, 0 to remove and 5 not upgraded.
Need to get 29.6 MB of archives.
After this operation, 243 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 mysql-common all 5.8+1.1.0build1 [6746 B]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 mysql-client-core-8.0 amd64 8.0.43-0ubuntu0.24.04.1 [2740 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 mysql-client-8.0 amd64 8.0.43-0ubuntu0.24.04.1 [22.4 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libevent-pthreads-2.1-7t64 amd64 2.1.12-stable-9ubuntu2 [7982 B]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libmecab2 amd64 0.996-14ubuntu4 [201 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libprotobuf-lite32t64 amd64 3.21.12-8.2ubuntu0.2 [238 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 mysql-server-core-8.0 amd64 8.0.43-0ubuntu0.24.04.1 [17.5 MB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 mysql-server-8.0 amd64 8.0.43-0ubuntu0.24.04.1 [1439 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libhtml-tagset-perl all 3.20-6 [11.3 kB]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 liburi-perl all 5.27-1 [85.0 kB]
Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libhtml-parser-perl amd64 3.81-1build3 [85.8 kB]
Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libbcbi-pm-perl all 4.63-1 [185 kB]
Get:13 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libfcgi0t64 amd64 2.4.2-2.1ubuntu0.24.04.1 [27.0 kB]
Get:14 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libfcgi-perl amd64 0.82+ds-3build2 [21.7 kB]
Get:15 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libbcbi-fast-perl all 1.2-17-1 [10.3 kB]
Get:16 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libclone-perl amd64 0.46-1build3 [10.7 kB]
Get:17 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libcode-locale-perl all 1.05-3 [11.6 kB]
Get:18 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libfcgi-bin amd64 2.4.2-2.1ubuntu0.24.04.1 [11.2 kB]
Get:19 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libhtml-template-perl all 2.97-2 [60.2 kB]
Get:20 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libtimedate-perl all 2.3300-2 [34.0 kB]
Get:21 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libhttp-date-perl all 6.06-1 [10.2 kB]
Get:22 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libio-html-perl all 1.004-3 [15.9 kB]
Get:23 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 liblwp-mediatypes-perl all 6.04-2 [20.1 kB]
Get:24 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libhttp-message-perl all 6.45-1ubuntu1 [78.2 kB]
Get:25 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 mecab-utils amd64 0.996-14ubuntu4 [4804 B]
Get:26 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 mecab-ipadic all 2.7.0-20070801+main-3 [6718 kB]
Get:27 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 mecab-ipadic-utf8 all 2.7.0-20070801+main-3 [4384 B]
Get:28 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 mysql-server all 8.0.43-0ubuntu0.24.04.1 [9520 B]
Fetched 29.6 MB in 0s (70.0 MB/s)
Preconfiguring packages ...
Selecting previously unselected package mysql-common.
(Reading database ... 71766 files and directories currently installed.)
Preparing to unpack .../0-mysql-common-5.8+1.1.0build1_all.deb ...
Unpacking mysql-common (5.8+1.1.0build1) ...
Selecting previously unselected package mysql-client-core-8.0.
Preparing to unpack .../1-mysql-client-core-8.0-8.0.43-0ubuntu0.24.04.1_amd64.deb ...
Unpacking mysql-client-core-8.0 (8.0.43-0ubuntu0.24.04.1) ...
Selecting previously unselected package mysql-client-8.0.
Preparing to unpack .../2-mysql-client-8.0-8.0.43-0ubuntu0.24.04.1_amd64.deb ...
Unpacking mysql-client-8.0 (8.0.43-0ubuntu0.24.04.1) ...
Selecting previously unselected package libevent-pthreads-2.1-7t64:amd64.
Preparing to unpack .../3-libevent-pthreads-2.1-7t64:amd64 (2.1.12-stable-9ubuntu2) ...
Selecting previously unselected package libmecab2:amd64.
Preparing to unpack .../4-libmecab2_0.996-14ubuntu4_amd64.deb ...
```

Step 5: Install PHP and FPM

```
# to check the PHP version
php --version
```

```
# to install php-fpm
sudo apt install php8.3-fpm
```

```
ubuntu@ip-172-31-42-153: ~
ubuntu@ip-172-31-42-153:~$ php --version
Command 'php' not found, but can be installed with:
sudo apt install php8.3-cli # version 8.3.6-0ubuntu0.24.04.3, or
sudo apt install php-cli # version 2:8.2+93ubuntu1
sudo apt install php8.2-cli # version 8.2.12-1ubuntu2
ubuntu@ip-172-31-42-153:~$ sudo apt install php8.3-fpm
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  php-common php8.3-cli php8.3-common php8.3-opcache php8.3-readline
Suggested packages:
  php-pear
The following NEW packages will be installed:
  php-common php8.3-cli php8.3-common php8.3-fpm php8.3-opcache php8.3-readline
0 upgraded, 6 newly installed, 0 to remove and 5 not upgraded.
Need to get 4979 kB of archives.
After this operation, 22.5 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 php-common all 2:93ubuntu2 [13.9 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-common amd64 8.3.6-0ubuntu0.24.04.5 [740 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-opcache amd64 8.3.6-0ubuntu0.24.04.5 [371 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-readline amd64 8.3.6-0ubuntu0.24.04.5 [13.5 kB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-cli amd64 8.3.6-0ubuntu0.24.04.5 [1915 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 php8.3-fpm amd64 8.3.6-0ubuntu0.24.04.5 [1925 kB]
Fetched 4979 kB in 0s (59.7 MB/s)
Selecting previously unselected package php-common.
(Reading database ... 72382 files and directories currently installed.)
Preparing to unpack .../0-php-common_2%3a93ubuntu2_all.deb ...
Unpacking php-common (2:93ubuntu2) ...
Selecting previously unselected package php8.3-common.
Preparing to unpack .../1-php8.3-common_8.3.6-0ubuntu0.24.04.5_amd64.deb ...
Unpacking php8.3-common (8.3.6-0ubuntu0.24.04.5) ...
Selecting previously unselected package php8.3-opcache.
Preparing to unpack .../2-php8.3-opcache_8.3.6-0ubuntu0.24.04.5_amd64.deb ...
Unpacking php8.3-opcache (8.3.6-0ubuntu0.24.04.5) ...
Selecting previously unselected package php8.3-readline.
Preparing to unpack .../3-php8.3-readline_8.3.6-0ubuntu0.24.04.5_amd64.deb ...
Unpacking php8.3-readline (8.3.6-0ubuntu0.24.04.5) ...
Selecting previously unselected package php8.3-cli.
Preparing to unpack .../4-php8.3-cli_8.3.6-0ubuntu0.24.04.5_amd64.deb ...
Unpacking php8.3-cli (8.3.6-0ubuntu0.24.04.5) ...
Selecting previously unselected package php8.3-fpm.
Preparing to unpack .../5-php8.3-fpm_8.3.6-0ubuntu0.24.04.5_amd64.deb ...
Unpacking php8.3-fpm (8.3.6-0ubuntu0.24.04.5) ...
Setting up php-common (2:93ubuntu2) ...
Created symlink /etc/systemd/system/timers.target.wants/phpsessionclean.timer → /usr/lib/systemd/system/phpsessionclean.timer.
Setting up php8.3-common (8.3.6-0ubuntu0.24.04.5) ...
Creating config file /etc/php/8.3/mods-available/calendar.ini with new version
Creating config file /etc/php/8.3/mods-available/ctype.ini with new version
Creating config file /etc/php/8.3/mods-available/exif.ini with new version
Creating config file /etc/php/8.3/mods-available/fileinfo.ini with new version
Creating config file /etc/php/8.3/mods-available/ffi.ini with new version
Creating config file /etc/php/8.3/mods-available/ftp.ini with new version
Creating config file /etc/php/8.3/mods-available/gettext.ini with new version
```

Step 6: Start, enable, and check the status of Nginx, MySQL, and PHP-FPM.

```
# to start
sudo systemctl start nginx mysql php8.3-fpm
```

```
# to enable
sudo systemctl enable nginx mysql php8.3-fpm
```

```
# to check the status
sudo systemctl status nginx mysql php8.3-fpm
```

```

ubuntu@ip-172-31-42-153: ~
ubuntu@ip-172-31-42-153:~$ sudo systemctl start nginx mysql php8.3-fpm
ubuntu@ip-172-31-42-153:~$ sudo systemctl enable nginx mysql php8.3-fpm
Synchronizing state of nginx.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable nginx
Synchronizing state of mysql.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable mysql
Synchronizing state of php8.3-fpm.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable php8.3-fpm
ubuntu@ip-172-31-42-153:~$ sudo systemctl status nginx mysql php8.3-fpm
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Sat 2025-08-23 19:35:59 UTC; 9min ago
     Docs: man:nginx(8)
    Main PID: 1806 (nginx)
      Tasks: 3 (limit: 1008)
     Memory: 2.4M (peak: 5.3M)
        CPU: 27ms
    CGroup: /system.slice/nginx.service
            └─1806 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
            └─1808 "nginx: worker process"
            └─1809 "nginx: worker process"

Aug 23 19:35:59 ip-172-31-42-153 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Aug 23 19:35:59 ip-172-31-42-153 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.

● mysql.service - MySQL Community Server
   Loaded: loaded (/usr/lib/systemd/system/mysql.service; enabled; preset: enabled)
   Active: active (running) since Sat 2025-08-23 19:37:49 UTC; 8min ago
     Main PID: 2577 (mysqld)
    Status: "Server is operational"
      Tasks: 37 (limit: 1008)
     Memory: 351.6M (peak: 378.2M)
        CPU: 4.311s
    CGroup: /system.slice/mysql.service
            └─2577 /usr/sbin/mysqld

Aug 23 19:37:48 ip-172-31-42-153 systemd[1]: Starting mysql.service - MySQL Community Server...
Aug 23 19:37:49 ip-172-31-42-153 systemd[1]: Started mysql.service - MySQL Community Server.

● php8.3-fpm.service - The PHP 8.3 FastCGI Process Manager
   Loaded: loaded (/usr/lib/systemd/system/php8.3-fpm.service; enabled; preset: enabled)
   Active: active (running) since Sat 2025-08-23 19:42:57 UTC; 2min 53s ago
     Docs: man:php-fpm8.3(8)
    Main PID: 9052 (php-fpm8.3)
    Status: "Processes active: 0, idle: 2, Requests: 0, slow: 0, Traffic: 0req/sec"
      Tasks: 3 (limit: 1008)
     Memory: 8.1M (peak: 8.7M)
        CPU: 61ms
    CGroup: /system.slice/php8.3-fpm.service
            └─9052 "php-fpm: master process /etc/php/8.3/fpm/php-fpm.conf"
            └─9053 "php-fpm: pool www"
            └─9054 "php-fpm: pool www"

Aug 23 19:42:57 ip-172-31-42-153 systemd[1]: Starting php8.3-fpm.service - The PHP 8.3 FastCGI Process Manager...
Aug 23 19:42:57 ip-172-31-42-153 systemd[1]: Started php8.3-fpm.service - The PHP 8.3 FastCGI Process Manager.
ubuntu@ip-172-31-42-153:~$

```

Step 7: Navigate to Nginx's default directory and create the files index.html and index.php.

```
# to navigate to default directory
sudo /var/www/html
```

```
# to create file
sudo vim file_name
```

```
# to view the code
cat file_name
```

```
#to restart and to verify the applied changes
sudo systemctl restart nginx mysql php8.3-fpm
```

```
ubuntu@ip-172-31-42-153: /var/www/html
ubuntu@ip-172-31-42-153:~$ cd /var/www/html/
ubuntu@ip-172-31-42-153:/var/www/html$ sudo vim index.html
ubuntu@ip-172-31-42-153:/var/www/html$ cat index.html
<h1> This is my LEMP website</h1>
ubuntu@ip-172-31-42-153:/var/www/html$ sudo vim index.php
ubuntu@ip-172-31-42-153:/var/www/html$ cat index.php
<?php
    phpinfo();
?>
ubuntu@ip-172-31-42-153:/var/www/html$ sudo systemctl restart nginx mysql php8.3-fpm
ubuntu@ip-172-31-42-153:/var/www/html$ |
```

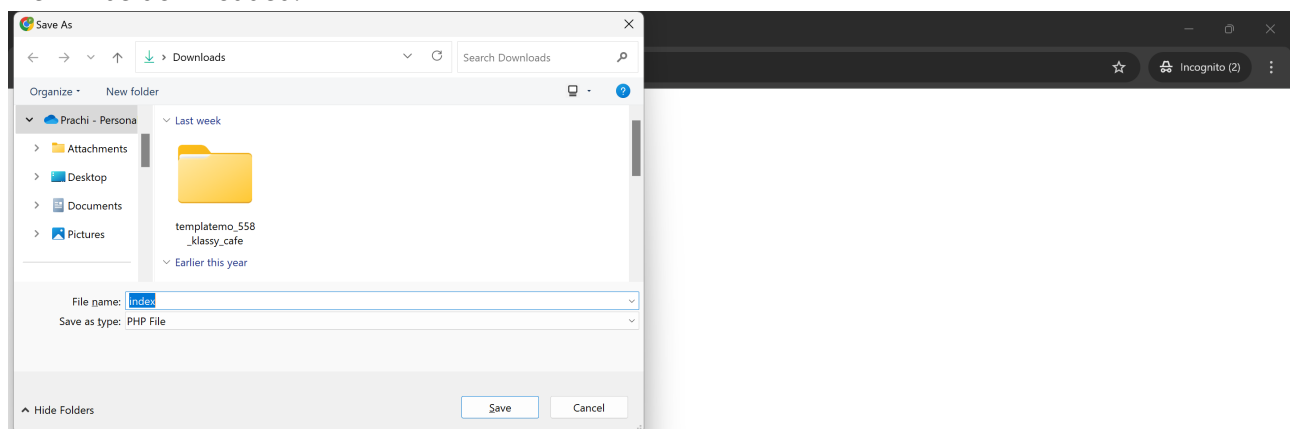
Step 7: Copy the public IP address and paste it into the browser to access the website.

- The index.html file is the default page.



This is my LEMP website

- To view the PHP page, enter publicip/index.php in the URL bar, but note that instead of displaying, the file will be downloaded.

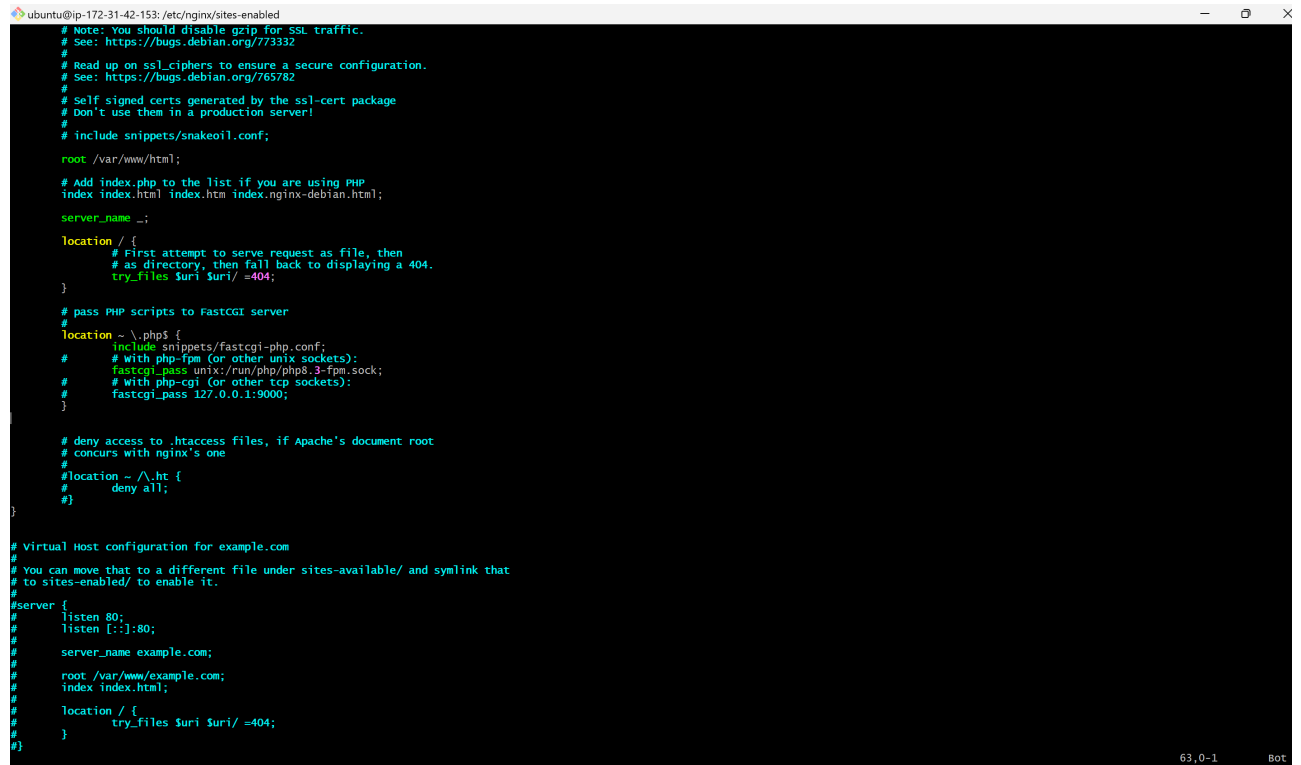


Step 8: Edit the Nginx default site configuration to prevent the PHP file from being downloaded and display the index.php page in the browser.

```
#to navigate nginx configuration file path
cd /etc/nginx/sites-enabled/default/
```

```
#to open default file
sudo vim default
```

- In the default file, locate the second location block (currently commented), uncomment the necessary lines, and update the PHP version to the current version within the same block.



```
ubuntu@ip-172-31-42-153: /etc/nginx/sites-enabled
# Note: You should disable gzip for SSL traffic.
# See: https://bugs.debian.org/773332
#
# Read up on ssl_ciphers to ensure a secure configuration.
# See: https://bugs.debian.org/765782
#
# Self signed certs generated by the ssl-cert package
# Don't use them in a production server!
#
# include snippets/snakeoil.conf;

root /var/www/html;

# Add index.php to the list if you are using PHP
index index.html index.htm index.nginx-debian.html;

server_name _;

location / {
    # First attempt to serve request as file, then
    # as directory, then fall back to displaying a 404.
    try_files $uri $uri/ =404;
}

# pass PHP scripts to FastCGI server
#
location ~ \.php$ {
    include snippets/fastcgi-php.conf;
    # with php-fpm (or other unix sockets):
    fastcgi_pass unix:/run/php/php8.3-fpm.sock;
    # with php-cgi (or other tcp sockets):
    fastcgi_pass 127.0.0.1:9000;
}

# deny access to .htaccess files, if Apache's document root
# concurs with nginx's one
#
#location ~ /\.ht {
#    deny all;
#}

# Virtual Host configuration for example.com
#
# You can move that to a different file under sites-available/ and symlink that
# to sites-enabled/ to enable it.

#server {
#    listen 80;
#    listen [::]:80;
#
#    server_name example.com;
#
#    root /var/www/example.com;
#    index index.html;
#
#    location / {
#        try_files $uri $uri/ =404;
#    }
#}
```

```
#to restart and to verify the applied changes
sudo systemctl restart nginx mysql php8.3-fpm
```

```
ubuntu@ip-172-31-42-153: /etc/nginx/sites-enabled
ubuntu@ip-172-31-42-153:/etc/nginx/sites-enabled$ cd /etc
ubuntu@ip-172-31-42-153:/etc$ cd nginx/
ubuntu@ip-172-31-42-153:/etc/nginx$ ls
conf.d      fastcgi_params  koi-win      modules-available  nginx.conf  scgi_params  sites-enabled  uwsgi_params
fastcgi.conf  koi-utf        mime.types   modules-enabled    proxy_params  sites-available  snippets      win-utf
ubuntu@ip-172-31-42-153:/etc/nginx$ cd sites-enabled/
ubuntu@ip-172-31-42-153:/etc/nginx/sites-enabled$ ls
default
ubuntu@ip-172-31-42-153:/etc/nginx/sites-enabled$ sudo vim default
ubuntu@ip-172-31-42-153:/etc/nginx/sites-enabled$ sudo systemctl restart nginx mysql php8.3-fpm
ubuntu@ip-172-31-42-153:/etc/nginx/sites-enabled$ |
```

Step 9: Verify that index.php is correctly deployed by accessing it via the server’s IP in a browser.

PHP 8.3.6 - phpinfo()

Not secure 54.196.37.250/index.php

Incognito

PHP Version 8.3.6



System	Linux ip-172-31-42-153 6.14.0-1011-aws #11-24.04.1-Ubuntu SMP Fri Aug 1 02:07:25 UTC 2025 x86_64
Build Date	Jul 14 2025 18:30:55
Build System	Linux
Server API	FPM/FastCGI
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc/php/8.3/fpm
Loaded Configuration File	/etc/php/8.3/fpm/php.ini
Scan this dir for additional .ini files	/etc/php/8.3/fpm/conf.d
Additional .ini files parsed	/etc/php/8.3/fpm/conf.d/10-opcache.ini, /etc/php/8.3/fpm/conf.d/10-pdo.ini, /etc/php/8.3/fpm/conf.d/20-calendar.ini, /etc/php/8.3/fpm/conf.d/20-ctype.ini, /etc/php/8.3/fpm/conf.d/20-exif.ini, /etc/php/8.3/fpm/conf.d/20-fi.ini, /etc/php/8.3/fpm/conf.d/20-fileinfo.ini, /etc/php/8.3/fpm/conf.d/20-ftp.ini, /etc/php/8.3/fpm/conf.d/20-gettext.ini, /etc/php/8.3/fpm/conf.d/20-iconv.ini, /etc/php/8.3/fpm/conf.d/20-phar.ini, /etc/php/8.3/fpm/conf.d/20-posix.ini, /etc/php/8.3/fpm/conf.d/20-readline.ini, /etc/php/8.3/fpm/conf.d/20-shmop.ini, /etc/php/8.3/fpm/conf.d/20-sockets.ini, /etc/php/8.3/fpm/conf.d/20-sysvmsg.ini, /etc/php/8.3/fpm/conf.d/20-sysvsem.ini, /etc/php/8.3/fpm/conf.d/20-tokenizer.ini
PHP API	20230831
PHP Extension	20230831
Zend Extension	420230831
Zend Extension Build	API420230831.NTS
PHP Extension Build	API20230831.NTS
Debug Build	no
Thread Safety	disabled
Zend Signal Handling	enabled
Zend Memory Manager	enabled
Zend Multibyte Support	disabled
Zend Max Execution Timers	disabled
IPv6 Support	enabled
DTrace Support	disabled
Registered PHP Streams	https, ftps, compress.zlib, php, file, glob, data, http, ftp, phar
Registered Stream Socket Transports	tcp, udp, unix, udg, ssl, tls, tlsv1.0, tlsv1.1, tlsv1.2, tlsv1.3
Registered Stream Filters	zlib.*, string.rot13, string.toupper, string.tolower, convert.*, consumed, dechunk, convert.iconv.*

Summary

This project provides a step-by-step process to set up a LEMP stack (Linux, Nginx, MySQL/MariaDB, PHP) on an Ubuntu EC2 instance. It covers installation, configuration, and deployment of dynamic PHP websites, including editing Nginx settings to properly serve index.php. By following these steps, users can host PHP-based applications efficiently on Nginx.